

CHENRUI ZHANG

☎ +1(509) 551-0982 ✉ chenrui151129@outlook.com 🎓 [Google Scholar](#) 📄 [ChenruiZhang](#) 🌐 [Chenrui](#)

EDUCATION

California State Polytechnic University, Pomona

Aug 2021 - Dec 2025

B.S. in Computer Science, Minor in Data Science

GPA: 3.9/4.0

Achievements: Cal-Bridge Scholar, President's Honor List 2021 - 2023

Relevant Coursework: Computer Vision, CUDA programming, Machine Learning

Incoming PhD Student

RESEARCH INTERESTS

Geometric deep learning, deep learning Theory, LLM

EXPERIENCE

Visiting Student – Wang Lab, UC San Diego

Mar 2025 – Present

Advisor: Dr. Yusu Wang

Project: AGOP Interpretability in Graph Neural Networks (Preparing submission for ICML 2026)

- Extended the **Average Gradient Outer Product (AGOP)** framework from MLPs to Graph Neural Networks to study how structural features influence model reasoning.
- Designed and implemented **AGOP-based importance analysis** to visualize feature-level and edge-level contributions, benchmarking against GNNExplainer and ground truth.
- Introduced a comparative evaluation using **adjacency-power metrics** to quantify alignment between AGOP explanations and structural ground truth.
- Explored architectural variants including **linear GCNs, ReLU-GCNs, and non-ReLU GCNs** to examine stability and gradient behavior under AGOP analysis.

Project: Structure-Aware Modeling and Set2Graph Framework

- Built a **synthetic data generation pipeline** using Gaussian Mixture Models to simulate detector events with controllable cluster centers, spreads, and activity states.
- Developed and compared **graph- and geometry-based architectures** for cluster center prediction, point-wise offset refinement, and adjacency learning, including **3D Graph UNet, Set2Graph**, and geometric transformer baselines.
- Designed training with dynamic- k regularization and multi-loss objectives (MAE, cosine similarity, L2) to jointly optimize center prediction and activity classification.
- Evaluated model generalization across varying cluster counts and size distributions, analyzing robustness and preservation of local geometric structure.

NSF-Sponsored Research Assistant – Mobility Scooter Lab

May 2023 – Jan 2025

Advisor: Dr. Tingting Chen

- Designed a multimodal representation pipeline by synchronizing pose keypoints and motion sensor data into a unified feature space, treating keypoints as nodes with temporal features.
- Implemented a **transformer encoder** to replace the LSTM-based model, using **self-attention** to capture global context across nodes and time, and applied a task-specific loss to refine feature embeddings before classification.
- Conducted **ablation studies** on embedding dimensions, attention heads, and positional encodings, demonstrating the performance advantage of the attention-based approach.
- Performed systematic **hyperparameter tuning** (learning rate, batch size, embedding size), improving feature representation quality and model generalization.
- Developed evaluation metrics (e.g., trunk lateral flexion angles) derived from keypoint configurations, confirming that learned embeddings preserved essential kinematic relationships.

Machine Learning Engineer Intern – Character Face Generation (CFG)

May 2024 – Sep 2024

Supervisor: Amrith Baskaran

- Reproduced the **GANFit pipeline** as a baseline and identified its limitations in recovering fine-grained facial geometry.
- Built a **mesh semantic segmentation model** based on the Mesh Transformer (MeT, 2023), constructing **triangle-, vertex-edge-, and Laplacian-graph representations** to encode both local and global topology.
- Extended MeT with an **edge-attention mechanism** emphasizing high-curvature and boundary edges, and integrated **edge-based pooling** for region-aware downsampling.
- Predicted **region-specific vertex offsets** to refine detailed structures (lips, eyelids) while simplifying flatter surfaces such as the forehead.
- Quantitatively evaluated results via **geodesic distances** and curvature statistics across poses and lighting, benchmarking against alternative configurations.

Supervisor: Dr. Hao Ji

- Evaluated students' work on CUDA programming, parallel algorithms, and GPU-based optimizations.
- Delivered lectures explaining the self-attention mechanism and its mathematical foundation, Covered applications in natural language processing and ViT.

SKILLS

- **Programming Languages:** Python, C/C++, Kotlin, Java, Bash
- **Frameworks and Libraries:** Pytorch, PyTorch Geometric, OpenCV, NumPy, CUDA, Pandas
- **Technologies and DB:** Git, Linux, Docker, REST APIs, MySQL

PUBLICATIONS

- [1] Thanh-Dat Nguyen, **Chenrui Zhang**, Melvin Gitbumrungsin, Amar Raheja, and Tingting Chen, “Remote Kinematic Analysis for Mobility Scooter Riders Leveraging Edge AI,” *Proceedings of the AAAI Symposium Series*, vol. 4, no. 1, pp. 314–318, Nov. 2024. [link]
- [2] Justin Chung, **Chenrui Zhang**, and Tingting Chen, “Mobility Scooter Riding Behavior Stability Analysis Based on Multimodal Contrastive Learning,” *IEEE International Conference on Big Data (BigData)*, pp. 6439–6445, Dec. 2024. [link]
- [3] Cleo Yau, **Chenrui Zhang**, and Tingting Chen, “A Machine Learning Powered Mobile Application for Mobility Scooter Driving Behavior Analysis,” *Fourth Annual Computer Science Conference for CSU Undergraduates*, Apr. 2024. [link]